The Action Is Not in the Workspace: An Agent's Tool Commitment Bypasses the Verbalizable Coordinates That Causally Reroute Its Answers

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Abstract

Anthropic [2026] show that language models carry an emergent, verbalizable *global workspace* ("J-space"): a few dozen concepts, read out by a Jacobian lens, whose coordinate interventions causally reroute multi-hop *answers* at mid-network depth. Their study contains no agentic content, and the public review of the paper explicitly names as unexamined whether an agent's *action* selection routes through this workspace or bypasses it. We test this on Qwen3.6-27B, an open-weights reasoning coding agent, using a row-restricted J-lens estimator whose readout directions we validate with a specificity control (a steer along the estimated direction flips a held answer to a designated alternative 17/20 at half the residual norm, while random and unrelated directions flip 0/20 and a random push of equal magnitude does not even move the answer). With the instrument validated, we find a **depth dissociation**: at mid-workspace (layers 27–43, ∼42–67% depth) the verbalizable direction *specifically* reroutes answers (11/20 vs random 0/20) but does *not* reroute the agent's tool commitment (edit vs bash: 0/60, *worse* than a random-direction control at 21/60); the action becomes specifically steerable only at the workspace→motor boundary (layers 47+, ≥73% depth, 60/60), and it is linearly readable in the same direction only there (AUROC peaks 0.74 at layer 47). Projecting out the top-10 verbalizable directions at the decision point leaves the commitment differential intact (0.174 → 0.167). We do not claim to discover the knowledge–action gap [Basu et al., 2026] nor that action is merely later than the answer in depth (both are known); the contribution is the *workspace dissociation*—the same intervention that reroutes the verbalizable answer in mid-network is bypassed by the action commitment—and the null ablation. The scoped implication: a J-lens-style verbalizable monitor reads a reasoning agent's answer but not its action commitment. Ledger, vectors, pre-registration, and a recompute-every-number evaluation are released; the readout curve recomputes offline to 10^{-6} (16/16 layers).

1 Introduction

A reasoning agent that decides to call an irreversible tool has, somewhere in its forward pass, committed to that action. Where? Anthropic [2026] give a sharp new vocabulary for "where": a Jacobian lens (J-lens) reveals an emergent *global workspace*, a small set of verbalizable concepts occupying a middle band of depth, whose coordinates can be swapped to causally change a model's multi-hop *answer*. Ablating the top workspace directions destroys multi-hop reasoning while sparing surface parsing. The paper studies pretrained-style completions; it contains no agent, tool-use, or action content, and the public review of the work notes that whether *action selection or tool routing bypasses the workspace* is a significant unexamined question for deployed agents.

Our prior arc supplies the exact complementary prior. For a long-horizon coding agent we showed the control of an action lives not at the mid-layer "task is done" verdict but in a late, task-matched commitment band ∼30 layers downstream [Vicentino, 2026a], that this late lever brakes irreversible actions [Vicentino, 2026b], and that a reasoning model's chain-of-thought is causal for its action only in that same late band [Vicentino, 2026c]. In the workspace vocabulary this predicts a specific, falsifiable claim: the agent's action commitment is *not* mid-workspace content; the knowledge–action gap is a workspace→motor transition. We test it directly.

Contribution. On one open-weights reasoning agent we establish a depth dissociation: the verbalizable-workspace direction that causally reroutes a held *answer* at mid-depth does *not* reroute the agent's *tool commitment* at the same depth (it is worse than a random control there), while both converge at the workspace→motor boundary; and ablating the verbalizable subspace at the decision point does not disturb the commitment. We are careful about novelty (§2): we neither discover the knowledge–action gap nor claim action is merely deeper than the answer—both are known. The new object is the *dissociation plus null ablation*, and its scoped monitoring implication.

2 Related work and novelty

A post-result literature scan returns no work testing whether an agent's action commitment routes through the *emergent* workspace; three neighbors bound our claims and are differentiated here. Basu et al. [2026] document a knowledge–action gap—internal probes separate cases the model fails to act on (98% vs 45%)—but treat it as an actionability limit; we do not claim the gap, we *localize* it as a workspace→motor transition and show the mid-workspace direction that reroutes the answer is bypassed by the action. Anonymous [2026a] read *whether* a tool is needed with a linear probe; we study *which* tool is committed and its causal routing, and show it is dissociated from the workspace direction and survives its ablation. Anonymous [2026b] decode tool-call dependency topology from the residual stream while explicitly disclaiming behavioral control, and Anonymous [2026c] show semantic direction forms early while deep layers stabilize outputs; because tool-related signals are linearly *readable* early, we scope our claim to the *specific causal steerability of the which-tool commitment via the verbalizable direction*, not readability in general. Global-workspace models of cognition in networks [Goyal et al., 2021, VanRullen & Kanai, 2021] *impose* a workspace as an architectural module; we test an *emergent* one and ask whether action routes through it.

3 Method

Model and decision points. Qwen3.6-27B (64 layers; depth% = $\ell/64$). From 99 SWE-bench Pro trajectories [Vicentino, 2026b] we take 60 deterministic `edit` (`str_replace_editor`) and 60 `bash` decision points, prefill-only, following the protocol of Vicentino [2026a]. Answers are a constructed 20-item few-shot two-hop QA set (single-token answers; 20/20 baseline accuracy).

Row-restricted J-lens estimator. For a single-token concept v and layer ℓ we estimate the readout row $g_{v,\ell} = \partial\,\mathrm{logit}_v(\text{final position})/\partial h_{\ell,\text{final}}$ by one backward pass per (v, prompt), parameters frozen and the embedding made the sole gradient leaf, averaged over an on-distribution estimation set. This is the diagonal (same-position) readout direction, not the full Jacobian matrix; we do not claim a complete reproduction of the J-lens.

Instrument validation (specificity control). The pre-registered coordinate *swap* $h \leftarrow h + V(\sigma(c) - c)$ is a near-no-op in this basis (the raw-gradient projections are near-equal, $c_{\text{correct}} \approx c_{\text{alt}}$), so we validate and use a *steer* along the normalized $(g_{\text{alt}} - g_{\text{correct}})$ direction scaled to $\alpha\|h\|$—a disclosed deviation to a second Anthropic-used intervention on the same directions. Table 1 shows the steer is specific: at α=0.5 it flips a held answer to the designated alternative 17/20, while unrelated-pair and random directions flip 0/20 and a random push of equal magnitude does not move the answer at all. (A readout-correlation check is uninformative here: RMSNorm is degree-0 homogeneous, so by Euler's theorem $g_v \cdot h \approx 0$ by construction.)

α	contrast→alt	unrelated→alt	random→alt	random moved off correct
0.5	**17/20**	0/20	0/20	0/20
1.0	**20/20**	0/20	0/20	19/20

Table 1: Specificity control: the estimated readout direction is causal and specific; equal-magnitude random pushes do not reroute (and at α=0.5 do not even move) the answer.

4 Results

H1: the action is readable only in the workspace tail. Figure 1 plots the AUROC of $(g_{\text{edit}} - g_{\text{bash}}) \cdot h$ separating edit- from bash-decision points by depth. It is at chance through mid-workspace and rises to a peak of 0.74 at layer 47 (73%), sustaining 0.59–0.69 into the motor band. The action's verbalizable readout emerges at the workspace tail, not mid-workspace. (The curve recomputes offline from the public vectors and residuals to 10^{-6} at all 16 layers.)

H2 and the dissociation: answers reroute in mid-workspace; the action does not. We steer the commitment along $(g_{\text{other}} - g_{\text{taken}})$ at the decision points (α=0.5), band-wise, against a random-direction control, and—parallel and apples-to-apples —steer the *answer* per band (Figure 2, Table 2). At mid-workspace the verbalizable direction *specifically* reroutes answers (11/20 vs random 0) but the action steer is *worse than random* (0/60 vs 21/60): no specific workspace-mediated rerouting of the commitment. Specific action control appears at the tail (60/60 vs random 23) and completes at motor. Because the identical magnitude at the identical layers reroutes the answer but not the action, the midW action result is not a magnitude/disruption artifact.

band	answer: contrast→alt (random)	action: edit→bash (random)
mid-workspace (L27–43)	**11/20 (0)**	**0/60 (21)**
tail (L47–55)	20/20 (0)	60/60 (23)
motor (L59–63)	20/20 (0)	60/60 (21)

Table 2: The dissociation. Same steer, same magnitude, same layers: the verbalizable direction reroutes the answer in mid-workspace but not the action.

H3: the commitment survives ablation of the verbalizable subspace. Projecting out the top-10 verbalizable directions (by lens activation on the agent distribution) at the decision points leaves the task-appropriate commitment differential essentially unchanged: $P(\text{edit})$ at edit points 0.474 → 0.473, at bash points 0.299 → 0.306; differential 0.174 → 0.167 (Figure 3). The

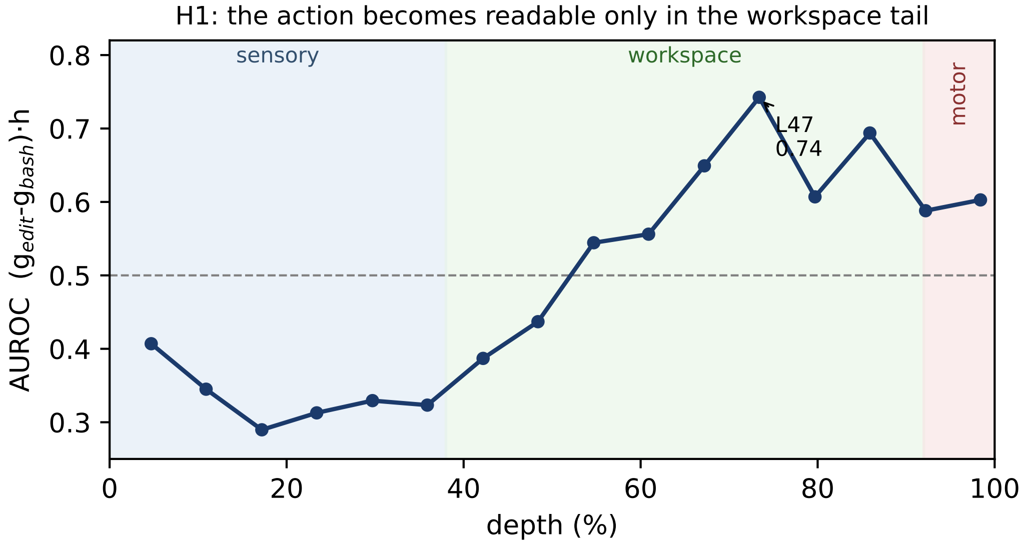

Figure 1: H1. The action's verbalizable readout (AUROC by depth) is at chance through mid-workspace and peaks at the tail (L47, 73%). Regimes after Anthropic [2026].

commitment does not live in the verbalizable subspace at the decision position—the load-bearing rebuttal to "you are re-reading an early-decodable tool signal."

5 Discussion

The verbalizable workspace that Anthropic [2026] steer to change an answer is, in this agent, bypassed by the action commitment: the same mid-network intervention that reroutes a held answer does not reroute the tool the agent commits to, which becomes causally steerable via the verbalizable direction only at the workspace→motor boundary, and survives ablation of the verbalizable subspace. This localizes the knowledge–action gap [Basu et al., 2026] as a workspace→motor depth transition and gives the arc's "lever is late" [Vicentino, 2026a,c] a mechanistic reading in workspace terms. The scoped implication for the workspace paper's own monitoring agenda: a J-lens-style verbalizable monitor reads a reasoning agent's answer/reasoning but not its action commitment—the monitor would see the model's stated intent and miss the committed act.

6 Limitations

One open-weights model, one behavior family (edit vs bash), n=60 action / 20 answer points, prefill-only, single magnitude α=0.5 (controlled within each band by the random baseline). The estimator is the row-restricted diagonal, not the full J-lens matrix, and we use a steer in place of the pre-registered coordinate swap (disclosed; the swap is a near-no-op in this basis; both are Anthropic-used and the steer is specificity-validated). Tool-related signals are linearly readable earlier than L47 in general [Anonymous, 2026b,a]; our claim is about specific causal steerability of the commitment via the verbalizable direction, not readability. No claim of architecture-generality; the workspace paper's own agent generality is untested.

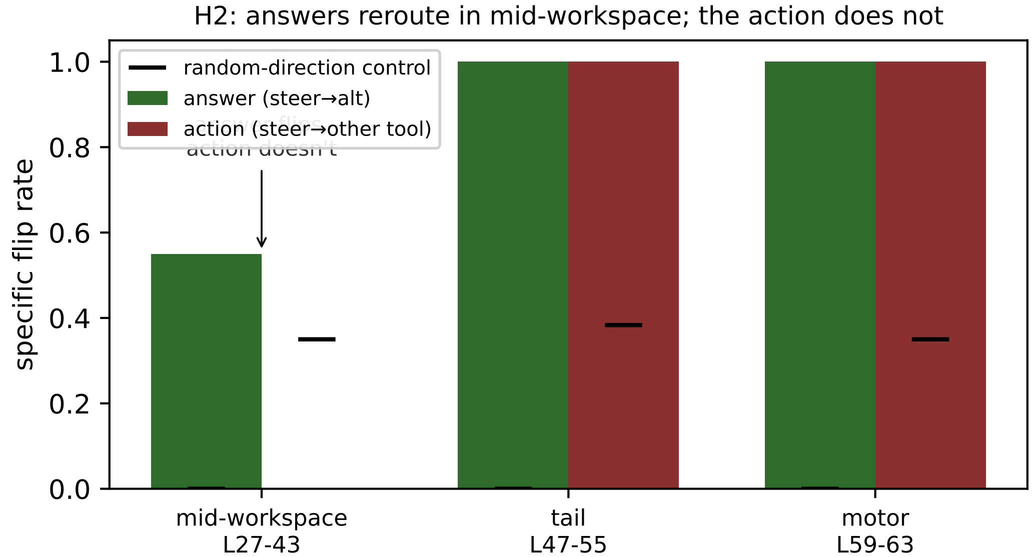

Figure 2: H2 dissociation. The verbalizable steer reroutes the *answer* in mid-workspace (11/20) but not the *action* (0/60, below the random control); both converge at the tail/motor. Black ticks: random-direction controls.

7 Reproducibility

Public HF ledger (`results/jspace_action.json`), estimated vectors, and decision-point residuals; harness `scripts/jspace_action.py` and pre-registration released. The evaluation `scripts/eval_jspace.py` recomputes the H1 readout offline from the public vectors and residuals (16/16 layers to 10^{-6}) and audits the dissociation logic (29/29). Steer and ablation numbers are model-dependent and were verified present in the public ledger step-by-step.

References

Anthropic Interpretability Team. A Global Workspace in Language Models. *Transformer Circuits Thread*, 2026. transformer-circuits.pub/2026/workspace.

S. Basu et al. Interpretability Without Actionability: The Knowledge–Action Gap in Language Models. *arXiv:2603.18353*, 2026.

LLM Agents Already Know When to Call Tools: Probing Tool-Necessity in Residual Streams. *arXiv:2605.09252*, 2026.

Tool-Call Dependency Structure is Linearly Decodable in LLM Agent Residual Streams. *arXiv:2605.25310*, 2026.

Do Agents Think Deeper? Layer-Wise Dynamics in Sequential Planning. *arXiv:2605.27935*, 2026.

A. Goyal et al. Coordination Among Neural Modules Through a Shared Global Workspace. *arXiv:2103.01197*, 2021.

R. VanRullen and R. Kanai. Deep Learning and the Global Workspace Theory. *Trends in Neurosciences*, 2021. arXiv:2012.10390.

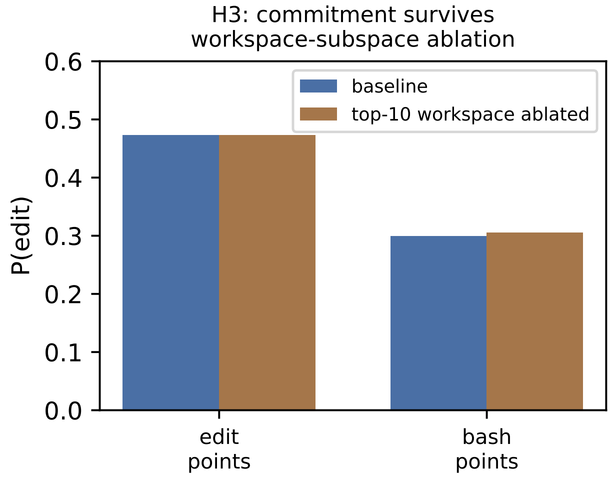

Figure 3: H3. Ablating the top-10 verbalizable directions at the decision point leaves the commitment differential intact.

C. Vicentino. The Lever Is Late: the Knowledge–Action Gap Is a Layer Gap. *Zenodo*, 2026. DOI 10.5281/zenodo.20534219.

C. Vicentino. The Commitment Lever Generalizes — and It Brakes. *Zenodo*, 2026. DOI 10.5281/zenodo.20679287.

C. Vicentino. The Late Channel: Chain-of-Thought Is Causal Only When It Changes the Outcome. *Zenodo*, 2026. DOI 10.5281/zenodo.20752895.